

Notes: Mitchell, Pulvino, and Stafford (JF, 2002)

Limited Arbitrage in Equity Markets

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1 Big picture

- **Question.** How costly and risky is arbitrage in real equity markets when an apparent law-of-one-price violation is simple to identify?
- **Empirical setting.** The paper studies negative-stub-value situations, where a parent firm's market value is less than the value of its publicly traded stake in a subsidiary.
- **Why this setting is powerful.** The mispricing is transparent: if the parent owns a stake in a listed subsidiary whose market value exceeds the entire parent's equity value, the parent's remaining stub assets are valued negatively by the market.
- **Main sample.** The authors identify 82 parent-subsidiary situations from 1985 to 2000 in which negative stub values appear.
- **Main empirical puzzle.** These situations look like arbitrage opportunities, but they often persist for long periods and sometimes terminate adversely.
- **Key finding 1.** About 30 percent of the sample does not converge before the economic link between parent and subsidiary is severed or otherwise eliminated.
- **Key finding 2.** A specialized arbitrageur earns high average returns, but the path of returns is volatile and exposes the arbitrageur to severe interim losses and margin risk.
- **Key finding 3.** Returns would be roughly 50 percent larger if convergence were smooth rather than as observed. Path risk, not only terminal risk, limits arbitrage.
- **Key frictions.** Short-sale costs, capital requirements, margin calls, incomplete convergence, and uncertainty about the distribution of arbitrage returns all matter.
- **Takeaway.** The law of one price does not enforce itself costlessly. Even apparently obvious mispricing can persist when arbitrage is capital-intensive, specialized, risky, and institutionally constrained.

2 Incremental contribution of the paper

2.1 Contribution relative to pure arbitrage theory

- Textbook arbitrage is riskless, requires no capital, and generates immediate profit.
- In practice, this paper shows that even highly transparent relative-value trades require capital, face short-sale frictions, and have uncertain holding periods.
- The paper therefore bridges the gap between the theoretical law of one price and the real-world business of arbitrage.

Interpretation: The incremental contribution is to show that arbitrage constraints are not only a theoretical possibility. They are visible in a setting where the mispricing is unusually easy to describe.

2.2 Contribution relative to limits-of-arbitrage models

- Earlier limits-of-arbitrage theories emphasize noise-trader risk, funding risk, margin constraints, and specialized arbitrage capital.
- This paper gives direct evidence on these mechanisms using actual equity-market mispricings.
- The sample includes both eventual convergence and adverse terminations, allowing the authors to measure terminal risk and path risk separately.
- The paper also quantifies how trading rules, leverage assumptions, and margin rules change returns.

Interpretation: The paper turns abstract limits-of-arbitrage forces into measurable objects: days invested, margin calls, short rebates, adverse terminations, and portfolio drawdowns.

2.3 Contribution relative to prior negative-stub studies

- Many discussions of negative stub values focus on famous Internet-era cases such as Creative Computers/Ubid or 3Com/Palm.
- This paper broadens the evidence to 82 situations over 1985–2000, covering many industries and years.
- It evaluates systematic trading rules rather than only single anecdotes.
- It distinguishes convergence caused by market prices from convergence caused by structural events, such as distributions, acquisitions, delistings, or termination of the parent-subsidiary link.

Interpretation: The paper’s incremental value is to move negative-stub arbitrage from anecdote to systematic evidence.

2.4 Contribution relative to empirical asset pricing

- Empirical asset pricing often infers mispricing from expected-return spreads or factor-model alphas.
- In this paper, the mispricing is directly observable from relative market values.
- This makes the setting unusually clean for testing why arbitrage capital does not instantly eliminate violations of relative value.

Interpretation: The paper separates two questions: whether mispricing exists and whether arbitrage capital can eliminate it. The first is unusually clear; the second is the empirical object of study.

3 Data and measurement

3.1 Negative stub values

Let the parent own a publicly traded stake in a subsidiary. Let

$$MV_{stake} \tag{1}$$

be the market value of the parent’s stake in the subsidiary, and let

$$MV_{parent\ equity} \tag{2}$$

be the market value of the parent’s equity. A simple stub value is

$$V_{stub} = MV_{parent\ equity} - MV_{stake}. \tag{3}$$

A negative stub arises when

$$V_{stub} < 0 \iff \frac{MV_{stake}}{MV_{parent\ equity}} > 1. \tag{4}$$

Interpretation: A negative stub means that the market values the parent’s other assets and liabilities together at a negative amount, after accounting for the market value of the subsidiary stake.

3.2 Two mispricing rules

The paper uses two rules for identifying and trading negative stub values.

Rule 1. The strict rule uses only the parent’s market value and the market value of its subsidiary stake:

$$V_{stub} < 0 \quad \text{if} \quad \frac{MV_{stake}}{MV_{parent \ equity}} > 1. \quad (5)$$

Rule 2. The broader rule incorporates the book value of parent equity:

$$V_{stub} < 0 \quad \text{if} \quad \frac{MV_{stake} + BV_{parent \ equity}}{MV_{parent \ equity}} > 1. \quad (6)$$

Interpretation: Rule 1 is cleaner and more conservative. Rule 2 catches cases where book equity gives additional information about the parent stub but may also include noisier situations.

3.3 Trading rule

The negative-stub trade is conceptually:

- short the subsidiary shares represented by the parent’s stake;
- buy the parent shares;
- wait for the relative-value discrepancy to close or for the parent-subsidiary link to be resolved.

A generic Rule 1 trading strategy is

$$\text{Buy if } \frac{MV_{stake}}{MV_{parent \ equity}} > \text{Buy Threshold}, \quad (7)$$

$$\text{Terminate if } \frac{MV_{stake}}{MV_{parent \ equity}} < \text{Sell Threshold}. \quad (8)$$

Interpretation: The buy threshold determines how extreme the mispricing must be before entering. The sell threshold determines when convergence is sufficiently complete to exit.

3.4 Costs and frictions

The paper considers several frictions that matter for actual arbitrage:

- brokerage commissions;
- short-sale rebates and negative rebates;
- capital requirements under textbook leverage, Regulation T leverage, and conservative leverage;
- maintenance margin requirements;
- risk-free interest earned on short proceeds;
- interim losses that force premature liquidation;
- the possibility that the parent-subsidiary relation is severed before convergence.

Interpretation: The paper is not simply calculating paper returns. It asks what a realistic arbitrageur could earn after accounting for the institutional structure of trading.

4 Empirical design and economic tests

4.1 Convergence and termination events

The first empirical question is whether the apparent mispricing eventually disappears. A deal can terminate because:

- the parent distributes subsidiary shares to its shareholders;
- a third party acquires the subsidiary;
- the parent acquires the remaining subsidiary shares;
- both parent and subsidiary are acquired;
- the parent or subsidiary is delisted;
- prices converge without a structural event.

Interpretation: This distinction is important. A trade can look profitable at entry but fail to converge if the subsidiary is acquired or the ownership link is severed before the market corrects relative prices.

4.2 Holding period risk

The paper measures the distribution of holding periods for different buy and sell thresholds. Holding periods range from a few days to several years.

Interpretation: A longer holding period increases exposure to margin risk, short-sale costs, opportunity costs, and the chance that capital providers lose patience.

4.3 Individual investment returns

For each trade, the paper computes annualized excess returns under different leverage rules:

- **textbook leverage:** assumes no initial margin requirement and no maintenance margin requirement;
- **Regulation T leverage:** accounts for regulatory initial and maintenance margins;
- **conservative leverage:** rules out positions that receive margin calls.

Interpretation: The gap between textbook and conservative returns measures how much of the apparent arbitrage profit depends on the ability to survive interim losses.

4.4 Calendar-time portfolio tests

The paper estimates factor regressions for portfolios of negative-stub investments:

$$R_{p,t} - R_{f,t} = a + b(R_{m,t} - R_{f,t}) + sSMB_t + hHML_t + e_t. \quad (9)$$

Interpretation: These regressions ask whether the trading strategy earns abnormal returns after controlling for standard equity risk factors.

4.5 Short-rebate evidence

The paper uses Ameritrade short-rebate data from 1998–2000. For some subsidiary shares, shorting is very costly, with large negative rebates.

Interpretation: Short-sale frictions are particularly relevant in negative-stub trades because the long leg is the parent and the short leg is often the subsidiary whose shares are hard to borrow.

5 Identification and interpretation

5.1 Why this is a clean limits-of-arbitrage setting

- The relative-value discrepancy is directly observable from market capitalization data.
- The parent and subsidiary have a transparent ownership link.
- A natural hedged trade exists: long parent, short subsidiary stake.
- The outcomes can be tracked until convergence or termination.

Interpretation: The paper does not need a complex asset-pricing model to infer mispricing. It can focus directly on why arbitrage does not remove the discrepancy.

5.2 Why arbitrage is limited

- Some trades do not converge before the economic link is severed.
- Interim price divergence creates margin calls.
- Short-sale rebates can be very negative, reducing returns.
- Arbitrageurs specializing in this strategy bear undiversified risk.
- The distribution of returns is hard to learn ex ante, limiting entry into the arbitrage business.

Interpretation: Mispricing persists because arbitrageurs must finance a risky dynamic strategy, not because they fail to notice the arithmetic.

5.3 Limits of the design

- Negative-stub situations are rare and may not represent all equity-market mispricings.
- Some assumptions are needed about the costs and leverage available to arbitrageurs.
- The paper observes realized paths, but would-be arbitrageurs at the time faced uncertainty over those paths.
- Rule 2 may include situations where book values introduce additional ambiguity.

Interpretation: The setting is unusually transparent, but actual arbitrage decisions still depend on ex ante beliefs, capital constraints, and the arbitrageur's institutional environment.

6 Tables and figures

The original visuals below are extracted directly from the paper. Adjacent figures and tables are placed on the same row where possible. If a row contains only one visual, it is shown at width 0.6.

6.1 Table I and Table II: sample and convergence

Read:

- Table I summarizes the distribution of negative-stub situations by industry and year.
- The Internet sector dominates the late 1990s part of the sample, but negative stubs also appear in capital goods, consumer, technology, transportation, and other sectors.
- Table II shows that under Rule 1, 48 of 66 deals have the mispricing eliminated, while 18 do not.
- Under Rule 2, 50 of 77 deals have the mispricing eliminated, while 27 do not.
- Adverse terminations include subsidiary acquisitions, parent acquisitions, and delistings.

Economic message: Mispricing often disappears, but not always. The risk that the ownership link is severed before convergence is a core limit to arbitrage.

Table I
Sample Summary

This table presents a summary of the negative-stub-value situations included in the sample. Rule 1 is used as the criterion for inclusion:

$$\text{Rule 1: } V_{\text{Stub}} < 0 \text{ if } \frac{MV_{\text{Stub}}}{MV_{\text{Parent Equity}}} > 1.0.$$

Panel A presents the frequency of situations by industry sector and year. Panel B presents negative-stub-value trades where either the parent or the subsidiary is an Internet firm. Internet firms are also included in the appropriate cells in Panel A. For situations that persist over many years, the year is determined by the date at which the stub value first goes negative. Numbers in each cell refer to subsidiaries/parents.

Industry	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
Panel A: Frequency of Negative-stub-value Situations by Industry Sector and Year																
Electronic materials			2/1	1/0	0/1			1/0	1/0			1/0		0/1		
Capital goods			2/1	1/0	0/1			1/0	1/0			1/0		0/1		
Consumer (cyclical)	1/0	2/2	1/0	1/0	3/2				0/1	0/1			0/1			
Consumer (non-cyc.)			3/2									2/1		1/1	2/3	
Chemical											1/0					
Health care										1/1					1/0	
Pharmaceuticals	0/1	1/2	0/3	0/2		0/1	0/2		0/1							
Telecommunications				2/1	1/1		1/0		1/1	1/1	1/1	1/1		0/1	5/6	2/1
Technology			1/1	0/1		1/0		1/1		1/0	4/3	1/1	2/1	8/7	3/3	
Transportation										0/1			1/1			
Utilities																0/1
Panel B: Frequency of "Internet" Negative-stub-value Situations by Year																
Internet											2/0		1/0	9/0	3/0	

Table I: sample summary

Table II
Frequency of Deals for Which Mispricing Is Eliminated

This table presents a summary of the frequency with which mispricings associated with negative-stub-value investments are eliminated at the time of deal termination. Results are presented using the following two mispricing criteria:

$$\text{Rule 1: } V_{\text{Stub}} < 0 \text{ if } \frac{MV_{\text{Stub}}}{MV_{\text{Parent Equity}}} > 1.0$$

$$\text{Rule 2: } V_{\text{Stub}} < 0 \text{ if } \frac{MV_{\text{Stub}} + BV_{\text{Parent Equity}}}{MV_{\text{Parent Equity}}} > 1.0$$

where V_{Stub} is the value of the parent's stub assets. Panel A describes the proportion of deals for which the mispricing is ultimately eliminated and Panel B describes the events that cause the mispricing to be eliminated.

Panel A: Frequency of Convergence for Negative-stub-value Investments			
Mispricing Measurement	Total Sample Size	Number (Percentage) of Deals for Which Mispricing Is Eliminated	Number (Percentage) of Deals for Which Mispricing Is Not Eliminated
Rule 1	66	48 (72.7%)	18 (27.3%)
Rule 2	77	50 (64.9%)	27 (35.1%)

Panel B: Description of Negative-stub-value Termination Events (Rule 1)			
Event	Number of Occurrences	Mispricing Eliminated	Mispricing Not Eliminated
Parent distributes subsidiary shares to parent shareholders	12	12	0
Third party acquires subsidiary	13	5	8
Parent acquires the subsidiary shares that it does not already own	5	3	2
Third party acquires both parent and subsidiary	7	4	3
Third party acquires parent	2	1	1
Subsidiary acquires parent	2	2	0
Parent stock is delisted	8	4	4
Both parent and subsidiary are delisted	2	2	0
Parent and subsidiary stock price changes eliminate mispricing	15	15	0
Mispricing not eliminated as of December 31, 2000	4	—	—
Total	70	48	18

Table II: frequency of deals for which mispricing is eliminated

6.2 Table III and Figure 1: holding periods and Creative Computers/Ubid

Read:

- Table III shows that negative-stub investments can last a long time.

- For Rule 1 and the broadest buy/sell threshold, the mean investment horizon for all investments is over 200 trading days.
- Some trades last more than 2,000 trading days.
- Figure 1 illustrates Creative Computers/Ubid, a famous case in which the subsidiary's value greatly exceeded the parent's market value.
- The figure shows that relative-value trades can become more extreme before converging.

Economic message: Holding-period risk is central. Even if ultimate convergence occurs, the arbitrageur must survive the path.

Table III
Investment Horizon

This table presents a summary of the number of days invested using two different trading strategies. Panels A and B show the number of days invested using Rule 1, for the full sample and for converged deals, respectively. Rule 1 is defined as:

$$\text{Place trade if } \frac{MV_{\text{Stake}}}{MV_{\text{Parent Equity}}} > \text{Buy Threshold}$$

and

$$\text{Terminate trade if } \frac{MV_{\text{Stake}}}{MV_{\text{Parent Equity}}} < \text{Sell Threshold.}$$

Panels C and D show the number of days invested using Rule 2 for the full sample and for converged deals, where Rule 2 is defined as:

$$\text{Place trade if } \frac{MV_{\text{Stake}} + BV_{\text{Parent Equity}}}{MV_{\text{Parent Equity}}} > \text{Buy Threshold}$$

$$\text{Terminate trade if } \frac{MV_{\text{Stake}} + BV_{\text{Parent Equity}}}{MV_{\text{Parent Equity}}} < \text{Sell Threshold.}$$

Converged deals are identified as those where the mispricing ratio is smaller at termination than at the initial date. Results are presented for various buy and sell thresholds. Table entries, other than buy and sell thresholds, represent trading days.

Buy Threshold	Sell Threshold	Number of Investments	Percentile					
			Minimum	25th	50th	75th	Maximum	Mean
Panel A: Number of Days Invested Using Rule 1 for All Investments								
1.00	0.8	110	1.0	29.0	92.0	274.0	2,796.0	236.3
1.25	1.0	75	2.0	28.5	84.0	213.0	2,413.0	214.4
1.50	1.0	39	13.0	92.5	159.0	258.3	2,370.0	343.1
Panel B: Number of Days Invested Using Rule 1 for Converged Investments								
1.00	0.8	79	3.0	25.0	74.0	207.5	1,804.0	176.5
1.25	1.0	40	2.0	26.5	66.5	127.5	1,792.0	164.7
1.50	1.0	20	13.0	46.0	117.5	163.0	1,792.0	265.9
Panel C: Number of Days Invested Using Rule 2 for All Investments								
1.00	0.8	130	3.0	46.0	155.5	380.0	2,818.0	309.7
1.25	1.0	119	3.0	41.5	114.0	285.0	2,818.0	236.0
1.50	1.0	67	7.0	68.8	182.0	336.0	2,804.0	327.7
Panel D: Number of Days Invested Using Rule 2 for Converged Investments								
1.00	0.8	94	3.0	35.0	109.0	279.0	2,818.0	271.1

Table III: investment horizon

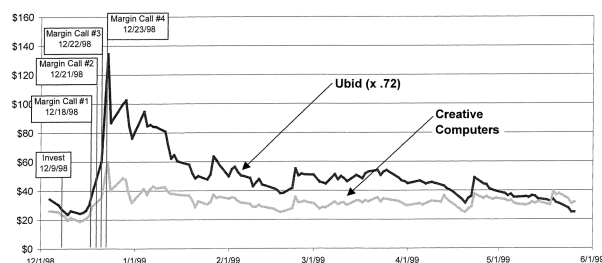


Figure 1: Paths of stock prices for Creative Computers and Ubid.

Figure 1: Creative Computers and Ubid prices

6.3 Table IV and Table V: individual and portfolio abnormal returns

Read:

- Table IV reports individual investment returns under different leverage assumptions.
- Textbook leverage produces high mean returns, but Regulation T and conservative leverage reduce performance materially.
- Conservative leverage returns are much smaller because they avoid strategies that trigger margin calls.
- Table V reports calendar-time portfolio regressions.
- Rule 1 portfolios generate positive abnormal returns, especially for moderate buy thresholds.
- Rule 2 portfolios have weaker abnormal performance.

Economic message: Apparent arbitrage returns are attractive, but realistic financing and risk constraints sharply reduce what can be captured.

Table IV
Individual Investment Returns Using Rule 1

This table summarizes annualized excess returns from investments in parent/subsidiary pairs using the following investment strategy:

$$\text{Place trade if } \frac{MV_{Stake}}{MV_{Parent\ Equity}} > \text{Buy Threshold}$$

and

$$\text{Terminate trade if } \frac{MV_{Stake}}{MV_{Parent\ Equity}} < \text{Sell Threshold.}$$

Returns are presented for various buy and sell thresholds and also for various leverage levels. Textbook leverage returns are calculated assuming Regulation T initial margin requirements but no maintenance margin requirements. Regulation T leverage returns are calculated using Regulation T initial and maintenance margin requirements: when margin calls are received, positions are partially liquidated such that maintenance margin requirements are satisfied. Conservative leverage returns are calculated using an Assets/Equity ratio that precludes margin calls for all parent/subsidiary pairs in the sample. All returns are annualized holding period returns in excess of the risk-free holding period return and assume that the risk-free rate is paid on short proceeds.

	Minimum Return (%)	Mean Return (%)	Maximum Return (%)	Investments With Negative Returns (%)	Investments That Receive Margin Calls (%)	Sample Size
Panel A: Buy Threshold = 1.0; Sell Threshold = 0.8						
All deals						
Textbook leverage	-104.9	18.5	120.0	17.3	—	110
Regulation T leverage	-104.1	12.7	120.0	22.7	23.6	110
Conservative leverage	-13.7	2.6	16.7	17.3	0.0	110
Converged deals						
Textbook leverage	-20.6	28.6	120.0	5.1	—	79
Regulation T leverage	-99.3	23.1	120.0	11.4	16.5	79
Conservative leverage	-6.1	8.2	35.7	5.1	0.0	79
Panel B: Buy Threshold = 1.25; Sell Threshold = 1.0						
All deals						
Textbook leverage	-103.9	21.6	71.6	16.0	—	75
Regulation T leverage	-104.1	13.8	71.6	21.3	26.7	75
Conservative leverage	-9.1	2.8	8.0	16.0	0.0	75
Converged deals						
Textbook leverage	-7.8	33.2	71.6	2.5	—	40
Regulation T leverage	-99.4	25.0	71.6	10.0	22.5	40
Conservative leverage	-2.0	8.6	17.2	2.5	0.0	40
Panel C: Buy Threshold = 1.5; Sell Threshold = 1.0						
All deals						
Textbook leverage	-103.9	24.2	71.6	20.5	—	39
Regulation T leverage	-104.1	10.1	71.6	35.9	38.5	39
Conservative leverage	-10.4	4.0	10.3	20.5	0.0	39
Converged deals	—	—	—	—	—	—

Table IV: individual investment returns using Rule 1

Table V
**Calendar-time Portfolio Regression Results
for the Negative-stub-value Investments
(January 1985 through December 2000)**

This table presents results from the following regression of monthly returns from a portfolio of negative-stub-value investments on common risk factors: $R_{p,t} - R_{f,t} = a + b(R_{m,t} - R_{f,t}) + sSMB_t + hHML_t + e_t$, where the dependent variable is the monthly return on a portfolio of negative-stub-value investments, R_p , in excess of the one-month Treasury Bill yield, R_f . The independent variables are the excess return of the market, $R_m - R_f$; the difference between a portfolio of "small" stocks and "big" stocks, SMB; and the difference between a portfolio of "high" book-to-market stocks and "low" book-to-market stocks, HML. See Fama and French (1993) for details on the construction of the factors. Results are presented for two investment strategies. Panel A shows the abnormal investment returns using Rule 1 defined as

$$\text{Place trade if } \frac{MV_{Stake}}{MV_{Parent\ Equity}} > \text{Buy Threshold}$$

$$\text{Terminate trade if } \frac{MV_{Stake}}{MV_{Parent\ Equity}} < \text{Sell Threshold.}$$

Panel B shows the abnormal investment returns using Rule 2, defined as

$$\text{Place trade if } \frac{MV_{Stake} + BV_{Parent\ Equity}}{MV_{Parent\ Equity}} > \text{Buy Threshold}$$

$$\text{Terminate trade if } \frac{MV_{Stake} + BV_{Parent\ Equity}}{MV_{Parent\ Equity}} < \text{Sell Threshold.}$$

Regression coefficients are presented for various buy and sell thresholds, with t -statistics in parentheses. The number of monthly portfolio returns are denoted by N .

Buy Threshold	Sell Threshold	No. of Investments	a	b	s	h	Adj. R^2 [N]	Annualized Abnormal Return
Panel A: Abnormal Investment Returns Using Rule 1								
1.00	0.8	110	0.825 (1.60)	-0.176 (-1.34)	-0.269 (-1.66)	0.088 (0.43)	0.050 [186]	9.9%
1.25	1.0	75	1.241 (2.17)	-0.173 (-1.18)	-0.137 (-0.78)	0.264 (1.19)	0.052 [172]	14.9%
1.50	1.0	39	0.514 (0.78)	-0.073 (-0.43)	-0.270 (-1.33)	0.284 (1.11)	0.047 [168]	6.2%
Panel B: Abnormal Investment Returns Using Rule 2								
1.00	0.8	130	0.021 (0.06)	0.120 (1.26)	-0.264 (-2.24)	0.183 (1.24)	0.061 [192]	0.2%
1.25	1.0	119	0.745 (1.54)	0.002 (0.01)	-0.076 (-0.49)	0.130 (0.68)	0.009 [192]	8.9%

Table V: calendar-time portfolio regression results

6.4 Figure 2 and Table VI: portfolio path risk

Read:

- Figure 2 plots the cumulative value of a negative-stub portfolio ignoring maintenance margin requirements.
- The portfolio grows substantially over time but suffers severe interim drawdowns, especially during the Internet bubble.
- The Creative Computers/Ubid episode and other spread-widening events create large temporary losses.
- Table VI shows that the pure negative-stub portfolio has high average returns but substantial volatility.
- Combining negative-stub investing with market and merger-arbitrage portfolios improves the Sharpe ratio.

Economic message: Diversification helps but does not eliminate funding risk. The strategy can have strong long-run returns and still be difficult to implement because of interim drawdowns.

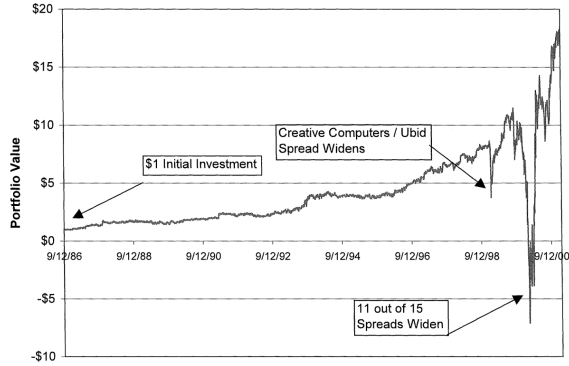


Figure 2. Daily portfolio values of negative-stub investments ignoring maintenance margin requirements. This figure shows the value of one dollar invested at the beginning of the period from 1986 to 2000 in negative-stub-value investments. Negative-stub-value portfolio values are obtained from an investment strategy based on Rule 1 (below) using a buy threshold of 1.25 and a sell threshold of 1.0:

$$\text{Place trade if } \frac{MV_{Stub}}{MV_{Parent\ Equity}} > 1.25 \text{ and Terminate trade if } \frac{MV_{Stub}}{MV_{Parent\ Equity}} < 1.0.$$

The portfolio value calculation cumulates daily payoffs from the arbitrage positions and ignores maintenance capital requirements, allowing the value of invested capital to become negative.

Figure 2: daily portfolio values ignoring maintenance margin

Table VI
Negative-stub-value Portfolio Return Series

This table presents the annual return series for the portfolio of negative-stub-value investments. Returns are presented for the pure negative-stub-value portfolio, the negative-stub-value portfolio combined with the market portfolio (41% negative stub values, 59% market), and for the negative-stub-value portfolio combined with a merger arbitrage portfolio (15% negative stub values, 85% merger arbitrage). Negative-stub-value portfolio returns are obtained from an investment strategy based on Rule 1 (below) using a buy threshold of 1.25 and a sell threshold of 1.0:

$$\text{Place trade if } \frac{MV_{Stub}}{MV_{Parent\ Equity}} > 1.25$$

and

$$\text{Terminate trade if } \frac{MV_{Stub}}{MV_{Parent\ Equity}} < 1.0.$$

Value-weighted CRSP returns are presented for comparison purposes. Merger arbitrage index returns are from Mitchell and Pulvino (2001). For the combination portfolios, weights are chosen to maximize the in-sample Sharpe ratio. Sharpe ratios are calculated using annual returns and annualized standard deviations. All annual returns are obtained by compounding monthly returns. Annualized standard deviations (Std) are obtained by multiplying the standard deviation of monthly returns by $\sqrt{12}$.

Year	Pure Negative-stub-value Portfolio	Negative-stub-value Portfolio Combined with the Market Portfolio	Negative-stub-value Portfolio Combined with the Merger Arbitrage Portfolio	Value-weighted CRSP Returns	Merger Arbitrage Index Return
1986	9.5%	1.4%	1.9%	15.6%	20.6%
1987	44.4%	21.4%	9.9%	1.8%	3.8%
1988	17.5%	17.7%	26.1%	17.6%	27.6%
1989	-0.3%	16.9%	4.8%	28.5%	5.4%
1990	29.3%	7.8%	7.9%	-6.0%	4.4%
1991	17.6%	27.3%	13.2%	33.6%	12.1%
1992	6.4%	8.2%	4.8%	9.0%	4.5%
1993	55.6%	28.3%	18.2%	11.5%	12.3%
1994	2.5%	1.0%	11.1%	-0.6%	12.6%
1995	2.1%	21.1%	9.7%	35.7%	11.0%
1996	41.1%	29.5%	19.1%	21.3%	15.4%
1997	20.5%	27.1%	13.1%	30.4%	11.6%
1998	-14.2%	9.3%	2.1%	22.5%	4.1%
1999	15.7%	26.5%	19.0%	24.9%	16.7%
2000	77.4%	20.6%	25.3%	-10.9%	16.6%
Mean	22.0%	17.5%	12.3%	14.4%	10.5%
Std	24.8%	12.3%	6.0%	15.4%	5.8%

Table VI: negative-stub portfolio return series

6.5 Table VII and Table VIII: short-sale frictions and news reactions

Read:

- Table VII uses Ameritrade short-rebate data to show that the subsidiary short leg can be expensive.
- Some stocks have very negative short rebates, such as Stratos Lightwave and Palm.
- These borrowing costs directly lower the profitability of negative-stub trades.
- Table VIII studies stock price reactions to news about distributions and tax status.
- Parent shares rise and subsidiary shares fall around favourable news, while net arbitrage positions earn positive announcement returns.

Economic message: Short-sale constraints and information events are both important. Arbitrageurs need to finance the short leg and bear the risk that convergence depends on uncertain corporate actions.

Table VII
Short-rebate Data

This table summarizes short-rebate data provided by Ameritrade Holding Corporation over the period December 1998 through October 2000. Panel A displays the number of sample firms with negative stub values between December 1998 and October 2000, the number covered by the Ameritrade short-rebate database, the number (and fraction) with negative short rebates, and the number (and fraction) with buy-ins. Panel B reports the minimum short-rebate transaction price paid for each subsidiary firm in the short-rebate database.

Panel A: Summary Statistics on Short-rebate Data						
	Rule 1			Rule 2		
	>1.0	>1.25	>1.5	>1.0	>1.25	>1.5
Number of firms with negative stub values between December 1998 and October 2000	28	21	17	32	31	23
Number of firms in short-rebate database	24	18	15	27	26	19
Number of firms in short-rebate database with negative short rebate	6	6	5	7	7	7
	(25.0%)	(33.3%)	(33.3%)	(25.9%)	(26.9%)	(36.8%)
Number of firms in short-rebate database with buy-ins	3	3	2	4	4	4
	(12.5%)	(16.7%)	(13.3%)	(14.8%)	(15.4%)	(21.1%)

Panel B: Minimum Short-Rebates Paid for Subsidiary Shares		
Company Name	Transaction Date	Minimum Short Rebate
Stratos Lightwave	July 6, 2000	-40.0%
Palm	July 28, 2000	-30.0%
Net2Phone	October 1, 1999	-8.0%
Retek	June 26, 2000	-4.0%
Plug Power	September 18, 2000	-4.0%
PFSWeb	January 20, 2000	-3.0%
MIPS Technology	September 28, 2000	-2.0%
Williams Communications	September 19, 2000	0.0%
Xpedior	December 30, 1999	0.0%
Iturf	September 24, 1999	0.0%
Ubid	January 26, 1999	0.0%
Marketwatch.Com	May 25, 1999	0.5%
Intimate Brands	February 23, 1999	0.5%
IXNet	September 7, 1999	1.0%
Interspeed	October 6, 1999	2.0%
Digex	August 16, 1999	2.0%
NetSilicon	September 30, 1999	2.3%
XM Satellite Radio	January 24, 2000	3.0%
US Search	March 28, 2000	3.0%
Veritas Software	June 7, 1999	3.0%
Barnes & Noble	August 23, 1999	4.0%
Kaiser Aluminum	December 22, 1998	4.4%
Nabisco Brands	July 19, 1999	4.7%
Keebler Foods	August 8, 2000	5.0%
CareInSite	December 16, 1999	5.0%
Superior Telecom	September 27, 2000	5.0%
Deltathree.Com	July 31, 2000	6.0%

Table VII: short-rebate data

Table VIII
Stock Price Reactions to the Release of News Concerning Distributions and Tax Status

This table presents the stock price reactions over the three days surrounding news announcements about spinoff distributions and IRS approval of these distributions as tax-free transactions. News announcements come from Dow Jones News Retrieval and the *Wall Street Journal*. Panel A describes the announcement period effects of all announcements by a single parent/subsidiary pair. Panel B describes the announcement period effects for firms where the prospectus indicates their intention to distribute the shares of the subsidiary to shareholders. Panel C describes the individual announcement period effects, treating multiple announcements by the same parent/subsidiary pair as distinct observations. Panel D describes the individual announcement period reactions to news, given that the mispricing ratio described by Rule 1 is greater than 1.0:

$$\text{Rule 1: } V_{\text{Stub}} < 0 \text{ if } \frac{MV_{\text{Stake}}}{MV_{\text{Parent Equity}}} > 1.0.$$

The t -statistics are calculated using the standard error of the mean. The p -values are calculated using the Wilcoxon signed rank test.

	Parent	Subsidiary	Net Position
Panel A: Effect of All News Announcements for a Given Firm			
Mean	6.17%	-2.25%	8.29%
t -statistic	2.64	-0.95	3.25
Median	3.88%	-2.76%	9.39%
p -value	0.0097	0.2553	0.0097
N	16.0	16	16
Panel B: Effect of All News Announcements for Firms Where Prospectus Indicates Intention to Distribute Shares			
Mean	4.13%	-4.18%	8.71%
t -statistic	1.39	-1.34	2.03
Median	3.86%	-6.91%	12.70%
p -value	0.1641	0.2031	0.0742
N	9	9	9
Panel C: All Announcements Treated Separately			
Mean	3.95%	-1.50%	5.30%
t -statistic	2.83	-0.96	3.11
Median	3.75%	-0.71%	3.31%
p -value	0.0074	0.3914	0.0021
N	25	24	25
Panel D: All Announcements Where the Mispricing Ratio Using Rule 1 Is Greater Than 1.0			
Mean	6.57%	-4.01%	11.37%
t -statistic	1.84	-1.07	2.29
Median	4.52%	-2.38%	10.85%

Table VIII: reactions to distribution and tax-status news

7 Short summary

- The paper studies 82 negative-stub-value situations between 1985 and 2000.
- Negative stubs occur when the parent's market value is less than the market value of its stake in a public subsidiary.
- These situations imply apparent arbitrage: buy the parent and short the subsidiary stake.
- Mispricing often persists for months or years.
- Around 30 percent of cases do not converge before the parent-subsidary link is severed or another adverse termination occurs.
- Trading-rule returns are positive on average but highly sensitive to leverage assumptions and margin requirements.
- Short-sale frictions, especially negative short rebates, can be large.
- Portfolio returns have large drawdowns, so funding risk matters even when average returns are high.
- News about distributions and tax treatment moves prices in the expected direction, confirming that the mispricing relates to the ownership link.
- The paper provides direct empirical evidence that arbitrage in equity markets is limited by capital, risk, and institutional frictions.

8 Conclusion

8.1 Core conclusion

Mitchell, Pulvino, and Stafford show that even obvious relative-value discrepancies in equity markets can persist. Negative-stub trades are easy to describe, but they are not riskless arbitrages. They require capital, involve uncertain holding periods, expose arbitrageurs to interim losses, and often depend on uncertain corporate events.

8.2 Economic intuition

An arbitrageur may be correct about fundamental relative value and still lose money if prices diverge before they converge, if borrowing costs on the short leg are high, or if the link between the parent and subsidiary is severed before convergence. This makes the arbitrage business risky and capital intensive.

8.3 Takeaway

The law of one price is enforced by real arbitrageurs who face financing constraints, short-sale frictions, and path risk. Because those constraints are meaningful, equity-market mispricing can survive even when it is transparent.